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CIS 344 Individual Project

Aquarium & Fish Care Store

An example of the mini world that I chose is an Aquarium & Fish Care Store. This store sells fish, aquariums, fish food, filters, and all other types of equipment used in the aquarium. This store offers people a chance to come and purchase all necessary items for taking care of their fish.

System Design Requirements

Requirements Gathering

To plan the database, I researched the basic needs of an aquarium and fish care store. I looked at the types of information a store needs to keep organized. An aquarium store needs to keep track of customers, fish, aquarium products, sales, and the items included in each sale. I used this information to decide what should be included in the database.

System Requirements

The database should be able to:

1. Track customer contact info and link them to orders.
2. Manage inventory and pricing for live fish and tank supplies (tanks, food, filters).
3. Track sales orders and line items per order.
4. Store each data category in its own table.

Based on these requirements, I decided that the database will use these main entities:

CUSTOMER: Stores information about customers.

FISH: Stores information about fish available in the store.

PRODUCT: Stores aquarium supplies and equipment.

SALE: Stores information about each purchase.

SALE_ITEM: Stores the items included in each sale and connects sales with items.

Customer: customer_id INT PK, first_name VARCHAR(40), last_name VARCHAR(40), email VARCHAR(80) UNIQUE, phone VARCHAR(15).

Fish: fish_id INT PK, species VARCHAR(40), water_type VARCHAR(15), price DECIMAL(10,2), stock_qty INT.

Product: product_id INT PK, prod_name VARCHAR(50), category VARCHAR(30), price DECIMAL(10,2), stock_qty INT.

Sale: sale_id INT PK, customer_id INT FK, sale_date DATE, total_amt DECIMAL(10,2).

SaleItem:

sale_item_id INT PK,

sale_id INT FK,

fish_id INT FK,

product_id INT FK,

quantity INT,

unit_price DECIMAL(10,2).

Foreign keys, Business Rule

One customer can place many sales, but each sale belongs to one customer (Sale.customer_id references Customer.customer_id). A sale contains multiple line items (SaleItem.sale_id references Sale.sale_id). SaleItem links to Fish.fish_id and Product.product_id to connect purchases to inventory.

Final Report

Challenges

I needed to ensure that all the tables are properly joined using foreign keys. Another challenge was ensuring that the database structure met the requirements of the aquarium store.
Outcome of the Project

Reflection

The Aquarium & Fish Care Store database was created successfully using MySQL Workbench. It comprises five related tables that include the customer table, fish table, product table, sales table, and sale item table. I have also drawn Chen and UML ER diagrams of the database structure.

Conclusion

This project has enabled me to understand how to implement requirements in a database using entities, primary keys, foreign keys, relationships, and SQL.